 

# ESKILSTUNA, Sweden

WMO: **024490**

Lat: **59.3825N** Lon: **16.4522E** Elev: **41** StdP: **14.67** Time Zone: **1.00 (EUC)** Period: **01-19** WBAN: **99999**

## Annual Heating, Humidification, and Ventilation Design Conditions

| | Coldest Month | Heating DB | | Humidification DP/MCDB and HR | | | | | | Coldest Month WS/MCDB | | | | MCWS/PCWD to 99.6% DB | | WSF | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 99.6% | | | 99% | | | 0.4% | | 1% | | | | | |
| | | 99.6% | 99% | DP | HR | MCDB | DP | HR | MCDB | WS | MCDB | WS | MCDB | MCWS | PCWD | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (1) | **1** | **-2.5** | **4.4** | **-4.7** | **4.3** | **-1.7** | **1.8** | **6.0** | **5.5** | **22.5** | **37.7** | **19.8** | **37.0** | **2.1** | **210** | **0.580** | (1) |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| | Hottest Month | Hottest Month DB Range | Cooling DB/MCWB | | | | | | Evaporation WB/MCDB | | | | | | MCWS/PCWD to 0.4% DB | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0.4% | | 1% | | 2% | | 0.4% | | 1% | | 2% | | | | |
| | | | DB | MCWB | DB | MCWB | DB | MCWB | WB | MCDB | WB | MCDB | WB | MCDB | MCWS | PCWD | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (2) | **7** | **19.3** | **81.4** | **64.5** | **77.9** | **63.5** | **74.8** | **62.0** | **67.6** | **75.3** | **65.6** | **73.7** | **63.8** | **72.0** | **6.7** | **250** | (2) |

| | Dehumidification DP/MCDB and HR | | | | | | | | | Enthalpy/MCDB | | | | | | Extreme Max WB | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0.4% | | | 1% | | | 2% | | | 0.4% | | 1% | | 2% | | | |
| | DP | HR | MCDB | DP | HR | MCDB | DP | HR | MCDB | Enth | MCDB | Enth | MCDB | Enth | MCDB | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (3) | **65.2** | **93.3** | **70.2** | **62.9** | **85.8** | **68.3** | **60.7** | **79.4** | **66.7** | **32.0** | **75.2** | **30.4** | **73.4** | **29.1** | **72.2** | **74.3** | (3) |

## Extreme Annual Design Conditions

| | Extreme Annual WS | | | | Extreme Annual Temperature | | | | n-Year Return Period Values of Extreme Temperature | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Mean | | Standard Deviation | | n=5 years | | n=10 years | | n=20 years | | n=50 years | | |
| | 1% | 2.5% | 5% | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| | (a) | (b) | (c) | | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (4) | **20.2** | **17.3** | **14.8** | DB | **-7.2** | **84.8** | **8.9** | **2.8** | **-13.6** | **86.8** | **-18.9** | **88.4** | **-23.9** | **90.0** | **-30.4** | **92.0** | (4) |
| (5) | | | | WB | **-4.7** | **69.5** | **8.6** | **2.3** | **-10.9** | **71.2** | **-15.9** | **72.6** | **-20.7** | **73.9** | **-26.9** | **75.6** | (5) |

## Monthly Climatic Design Conditions

| | | | | Annual | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (6) | Temperatures, Degree-Days and Degree-Hours | | DBAvg | **43.9** | **25.4** | **27.6** | **33.7** | **42.0** | **51.6** | **58.4** | **63.5** | **60.8** | **53.1** | **43.7** | **36.6** | **29.5** | (6) |
| (7) | | | DBStd | **14.81** | **10.16** | **9.50** | **7.60** | **6.13** | **6.53** | **5.38** | **4.81** | **4.75** | **5.64** | **6.89** | **7.07** | **9.93** | (7) |
| (8) | | | HDD50 | **3493** | **763** | **628** | **504** | **249** | **59** | **3** | **0** | **0** | **33** | **217** | **402** | **635** | (8) |
| (9) | | | HDD65 | **7763** | **1228** | **1048** | **969** | **691** | **417** | **208** | **89** | **145** | **356** | **661** | **851** | **1100** | (9) |
| (10) | | | CDD50 | **1273** | **0** | **0** | **0** | **8** | **108** | **255** | **417** | **336** | **127** | **21** | **1** | **0** | (10) |
| (11) | | | CDD65 | **69** | **0** | **0** | **0** | **0** | **1** | **10** | **41** | **16** | **1** | **0** | **0** | **0** | (11) |
| (12) | | | CDH74 | **850** | **0** | **0** | **0** | **0** | **39** | **165** | **491** | **152** | **3** | **0** | **0** | **0** | (12) |
| (13) | | | CDH80 | **158** | **0** | **0** | **0** | **0** | **1** | **23** | **115** | **19** | **0** | **0** | **0** | **0** | (13) |
| (14) | Wind | | WSAvg | **6.4** | **6.4** | **6.7** | **7.2** | **6.7** | **6.4** | **6.7** | **6.1** | **5.6** | **5.9** | **6.3** | **6.4** | **6.7** | (14) |
| (15) | Precipitation | | PrecAvg | **23.7** | **1.4** | **1.3** | **1.1** | **1.3** | **1.7** | **2.6** | **3.3** | **3.0** | **2.1** | **2.2** | **2.1** | **1.8** | (15) |
| (16) | | | PrecMax | **30.0** | **3.1** | **2.6** | **2.2** | **3.9** | **3.3** | **4.6** | **7.7** | **7.1** | **6.1** | **4.8** | **4.7** | **3.7** | (16) |
| (17) | | | PrecMin | **20.9** | **0.3** | **0.5** | **0.1** | **0.2** | **0.6** | **0.7** | **0.9** | **0.2** | **0.3** | **0.9** | **0.6** | **0.5** | (17) |
| (18) | | | PrecStd | **2.1** | **0.7** | **0.5** | **0.4** | **0.9** | **0.7** | **1.0** | **1.5** | **1.6** | **1.3** | **1.0** | **1.1** | **0.9** | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% | DB | | **44.0** | **47.3** | **56.6** | **68.1** | **77.9** | **83.0** | **87.3** | **82.2** | **73.5** | **62.1** | **53.8** | **50.8** | (19) |
| (20) | | | MCWB | | **41.5** | **43.5** | **46.8** | **51.9** | **59.2** | **63.6** | **65.2** | **66.6** | **61.8** | **56.0** | **51.3** | **47.7** | (20) |
| (21) | | 2% | DB | | **41.6** | **43.9** | **51.8** | **63.1** | **74.3** | **78.8** | **83.3** | **78.0** | **69.5** | **58.8** | **50.9** | **47.0** | (21) |
| (22) | | | MCWB | | **39.6** | **40.4** | **43.6** | **49.8** | **57.9** | **62.6** | **65.2** | **65.3** | **60.2** | **54.5** | **49.1** | **44.4** | (22) |
| (23) | | 5% | DB | | **39.5** | **41.7** | **47.7** | **58.9** | **70.1** | **74.6** | **79.5** | **74.7** | **65.8** | **56.6** | **48.5** | **43.6** | (23) |
| (24) | | | MCWB | | **38.0** | **39.3** | **41.7** | **48.0** | **56.3** | **60.5** | **64.3** | **63.4** | **58.4** | **53.1** | **46.8** | **41.4** | (24) |
| (25) | | 10% | DB | | **37.4** | **39.6** | **44.1** | **54.6** | **65.4** | **71.0** | **75.9** | **71.5** | **63.0** | **54.2** | **46.5** | **41.1** | (25) |
| (26) | | | MCWB | | **36.0** | **37.5** | **39.4** | **45.4** | **54.2** | **58.8** | **63.4** | **61.7** | **57.1** | **51.5** | **44.8** | **39.3** | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% | WB | | **42.3** | **44.9** | **48.4** | **53.3** | **62.8** | **66.6** | **70.8** | **70.2** | **63.6** | **57.8** | **51.9** | **48.4** | (27) |
| (28) | | | MCDB | | **43.4** | **46.3** | **54.7** | **64.4** | **74.5** | **77.4** | **77.6** | **77.2** | **71.0** | **60.4** | **53.2** | **50.3** | (28) |
| (29) | | 2% | WB | | **40.0** | **41.3** | **44.7** | **51.0** | **60.0** | **64.4** | **68.2** | **67.6** | **61.4** | **55.5** | **49.7** | **44.9** | (29) |
| (30) | | | MCDB | | **41.1** | **42.8** | **50.7** | **60.5** | **69.4** | **74.2** | **75.8** | **74.4** | **67.2** | **58.3** | **51.0** | **46.7** | (30) |
| (31) | | 5% | WB | | **38.3** | **39.6** | **42.5** | **49.0** | **57.5** | **62.2** | **66.2** | **65.2** | **59.6** | **53.8** | **47.4** | **41.9** | (31) |
| (32) | | | MCDB | | **39.5** | **41.3** | **46.9** | **57.5** | **67.4** | **72.3** | **74.3** | **72.0** | **64.4** | **56.1** | **48.6** | **43.3** | (32) |
| (33) | | 10% | WB | | **36.2** | **37.7** | **40.5** | **46.7** | **55.3** | **60.2** | **64.5** | **63.2** | **58.0** | **52.2** | **45.4** | **39.6** | (33) |
| (34) | | | MCDB | | **37.4** | **39.6** | **43.8** | **53.7** | **64.9** | **68.8** | **73.5** | **69.4** | **62.4** | **54.4** | **46.6** | **41.2** | (34) |
| (35) | Mean Daily Temperature Range | | MDBR | | **9.8** | **11.4** | **14.1** | **19.4** | **20.5** | **19.6** | **19.3** | **18.6** | **16.7** | **13.3** | **8.4** | **9.1** | (35) |
| (36) | | 5% DB | MCDBR | | **9.1** | **9.9** | **18.7** | **27.8** | **28.4** | **27.9** | **27.3** | **24.0** | **20.5** | **13.7** | **10.8** | **9.3** | (36) |
| (37) | | | MCWBR | | **8.3** | **7.9** | **12.5** | **16.7** | **15.7** | **14.0** | **12.7** | **12.3** | **12.2** | **10.0** | **9.7** | **9.1** | (37) |
| (38) | | 5% WB | MCDBR | | **8.4** | **9.0** | **17.4** | **24.5** | **24.7** | **23.3** | **21.7** | **20.9** | **18.5** | **12.6** | **10.7** | **9.7** | (38) |
| (39) | | | MCWBR | | **8.2** | **7.5** | **12.0** | **15.0** | **14.2** | **12.7** | **11.4** | **12.0** | **11.5** | **9.8** | **9.9** | **9.6** | (39) |
| (40) | Clear-Sky Solar Irradiance | taub | | | **0.276** | **0.317** | **0.338** | **0.375** | **0.371** | **0.362** | **0.387** | **0.379** | **0.345** | **0.334** | **0.303** | **0.260** | (40) |
| (41) | | taud | | | **2.209** | **2.275** | **2.353** | **2.316** | **2.393** | **2.441** | **2.391** | **2.408** | **2.491** | **2.422** | **2.269** | **2.140** | (41) |
| (42) | | Ebn at Noon | | | **166** | **219** | **253** | **262** | **272** | **277** | **266** | **259** | **251** | **213** | **153** | **125** | (42) |
| (43) | | Edh at Noon | | | **14** | **22** | **28** | **34** | **34** | **33** | **34** | **31** | **24** | **19** | **13** | **10** | (43) |
| (44) | All-Sky Solar Radiation | RadAvg | | | **100** | **301** | **781** | **1249** | **1622** | **1813** | **1658** | **1314** | **865** | **387** | **127** | **67** | (44) |
| (45) | | RadStd | | | **16** | **44** | **96** | **123** | **147** | **109** | **164** | **131** | **99** | **63** | **28** | **7** | (45) |

## Historical Trends

| | | DBAvg | Heating | | Cooling | | | Degree-Days | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 99% DB | 99% DP | 1% DB | 1% WB | 1% DP | HDD50 | HDD65 | CDD50 | CDD65 | |
| | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | |
| (46) | Station Only | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (46) |
| (47) | Regional (54 neighbors) | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **-218** | **-265** | **N/A** | **N/A** | (47) |

Nomenclature: See separate page